

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Software Engineering and Data Engineering professional with **5+ years of experience** building scalable data solutions, distributed processing workflows, data services, and analytics-driven platforms across enterprise environments.
- Strong programming expertise in **Python, Java, Scala, and SQL**, with hands-on experience developing reliable batch and real-time data pipelines for processing structured and semi-structured datasets.
- Experienced in **distributed data processing, Apache Spark, Apache Flink, Kafka, and large-scale data systems**, supporting high-volume workloads, analytical use cases, and downstream machine-learning applications.
- Strong background in **data modeling, ETL/ELT development, data transformation, schema design, data validation, and data quality**, enabling trusted datasets for analytics and business decision-making.
- Experienced in designing and debugging **data services and cloud-based data pipelines** using AWS, Airflow, Spark, relational databases, REST APIs, and event-driven architectures.
- Proven ability to collaborate with **engineering, product, business, analytics, and data science teams** to translate ambiguous requirements into scalable data products, metrics, and machine-learning data workflows.
- Strong focus on **elegant code, documentation, data debugging, observability, performance optimization, data quality, and independent problem solving** within rapidly changing Agile environments.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Query Languages:** Python, Java, Scala, SQL, C#, TypeScript, JavaScript, Bash, Object-Oriented Programming, Data Structures, Algorithms.
- **Big Data & Distributed Processing:** Apache Spark, PySpark, Spark SQL, Apache Flink, Hadoop, Kafka, distributed data processing, batch processing, stream processing, web-scale datasets.
- **Data Engineering & Modeling:** ETL/ELT Pipelines, Batch Data Pipelines, Real-Time Data Pipelines, Data Modeling, Dimensional Modeling, Data Transformation, Schema Design, Data Validation, Data Quality, Data Lineage.
- **Databases & Data Platforms:** PostgreSQL, MySQL, Oracle, SQL Server, Snowflake, Amazon S3, AWS Glue, AWS Lambda, Redshift, relational databases, data warehouses.
- **Orchestration & Cloud:** Apache Airflow, AWS, Docker, Kubernetes, REST APIs, Microservices, Event-Driven Architecture, Distributed Systems, Data Services, CI/CD Pipelines.
- **Analytics & Machine Learning Data:** Analytical Data Processing, Business Metrics, Feature Data Pipelines, ML Data Preparation, Data Aggregation, Data Profiling, Data Reconciliation, Statistical Data Validation.
- **Engineering & DevOps Tools:** Git, GitHub, GitLab, Jenkins, TeamCity, Maven, Jira, Confluence, Splunk, PyTest, JUnit, TestNG, Agile/Scrum, Monitoring, Data Debugging, Technical Documentation.

Software Engineer

Oct 2024 – Present

PCMI LLC (Full Time)

Chicago, IL, USA

- Designed and developed scalable **batch and real-time data pipelines** using Python, Java, SQL, Apache Spark, and Kafka to ingest, transform, validate, and process high-volume operational datasets supporting enterprise analytics and downstream applications.
- Implemented distributed data-processing workflows with **Apache Spark, PySpark, and Spark SQL**, optimizing partitioning, joins, caching, and transformations to efficiently process large datasets across distributed computing environments.
- Developed reusable **data models and transformation layers** for structured and semi-structured insurance and contract data, defining schemas, relationships, business rules, aggregations, and dimensional structures used for reporting and analytical workloads.
- Built event-driven data-processing services using **Kafka and REST APIs** to capture application events and operational changes, enabling near-real-time processing and providing reliable datasets for analytics and product functionality.
- Established automated **data quality and validation** controls using Python and SQL to identify missing values, duplicates, schema inconsistencies, reconciliation failures, and unexpected data anomalies before datasets reached downstream consumers.
- Developed and maintained production data workflows with **Apache Airflow, AWS S3, AWS Lambda, AWS Glue, and relational databases**, implementing scheduling, dependency management, retries, logging, monitoring, and failure recovery for reliable pipeline execution.
- Partnered with **product, software engineering, analytics, business, and data teams** to translate ambiguous requirements into scalable datasets, business metrics, data services, and data-processing solutions while independently driving technical implementation.
- Investigated complex production data issues using **SQL, Splunk, pipeline logs, and data profiling**, performing root-cause analysis across distributed services while maintaining technical documentation, data mappings, troubleshooting procedures, and operational runbooks.

Environments: Python, Java, Scala, SQL, Apache Spark, PySpark, Spark SQL, Apache Kafka, Apache Airflow, AWS S3, AWS Glue, AWS Lambda, PostgreSQL, REST APIs, ETL/ELT, Data Modeling, Distributed Systems, Docker, Kubernetes, Git, TeamCity, Splunk, Jira, Confluence, Agile/Scrum.

Software Engineer

Dec 2023 – Sept 2024

Client: CVS

Remote, USA

- Developed **Python, Java, and SQL-based data pipelines** for ingesting, transforming, and aggregating high-volume healthcare and application datasets used by operational reporting, analytical applications, and downstream data consumers.
- Implemented distributed data transformations using **Apache Spark and PySpark**, applying filtering, joins, aggregations, schema transformations, and partitioning strategies to efficiently process large structured and semi-structured datasets.
- Designed logical and dimensional **data models** to organize member, transaction, application, and operational data into reusable analytical datasets while maintaining consistent business definitions across downstream applications.
- Developed **real-time processing workflows** using Kafka and event-driven services to consume application events, validate incoming records, apply transformation logic, and make processed information available to downstream systems with low latency.
- Implemented comprehensive **data quality frameworks** using SQL and Python for completeness, uniqueness, referential integrity, reconciliation, schema validation, and business-rule checks, improving confidence in analytics and operational data products.
- Built and maintained cloud-based ETL/ELT workflows using **AWS Lambda, S3, Airflow, REST APIs, and relational databases**, adding automated scheduling, logging, retry mechanisms, error handling, and pipeline monitoring.
- Worked closely with **data science, engineering, product, and analytics teams** to curate datasets and reusable features for analytical insights and machine-learning workflows while responding effectively to evolving and ambiguous requirements.
- Performed production **data debugging and root-cause analysis** using SQL, Splunk, API logs, pipeline metrics, and source-to-target reconciliation, and documented data contracts, transformation rules, dependencies, and operational troubleshooting procedures.

Environments: Python, Java, SQL, Apache Spark, PySpark, Apache Kafka, Apache Airflow, AWS Lambda, AWS S3, PostgreSQL, MySQL, REST APIs, ETL/ELT, Data Modeling, Data Quality, Distributed Data Processing, Splunk, GitLab, Jenkins, Docker, Jira, Confluence, Agile/Scrum.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Developed backend data-processing components using **Java, Python, and SQL** to ingest, transform, validate, and deliver enterprise application data across SaaS modules and downstream analytical systems.
- Built and enhanced **ETL data pipelines** for processing operational datasets from relational databases, APIs, and application services, applying cleansing, transformation, enrichment, and aggregation rules before loading curated datasets.
- Utilized **Apache Spark and Spark SQL** for distributed processing of large datasets, implementing reusable transformation logic and optimizing queries, partitioning, and joins for improved processing efficiency.
- Created normalized and dimensional **data models** representing enterprise transactions, customers, application activity, and reference data while maintaining consistent schemas and relationships across source and downstream systems.
- Implemented automated **data quality checks** using SQL and Python to validate source-to-target mappings, record counts, field-level transformations, duplicates, null values, referential integrity, and business-rule compliance.
- Integrated application services and data workflows using **REST APIs, Kafka, and asynchronous processing patterns**, improving data movement between distributed components and supporting near-real-time operational requirements.
- Collaborated with **software engineers, product managers, analysts, and business stakeholders** during Agile development cycles to understand data requirements, investigate anomalies, and deliver reliable datasets and analytical capabilities.
- Performed detailed **data debugging and production troubleshooting** using SQL, application logs, pipeline execution details, and reconciliation queries while maintaining documentation for data mappings, schemas, technical workflows, and known failure scenarios.

Environments: Java, Python, Scala, SQL, Apache Spark, Spark SQL, Apache Kafka, ETL/ELT, Data Modeling, PostgreSQL, Oracle, MySQL, REST APIs, Distributed Systems, Jenkins, Git, Jira, Confluence, Docker, Linux, Agile/Scrum.

Associate Software Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed Java, Python, and **SQL-based data-processing utilities** to extract, validate, transform, and reconcile application data from relational databases and REST-based services supporting enterprise software platforms.
- Supported development and maintenance of **batch data pipelines** for transferring operational data between source applications, databases, and downstream systems while implementing transformation and business-validation rules.
- Worked with **Apache Spark and Spark SQL** to perform distributed transformations, filtering, aggregations, and analysis of large datasets while gaining practical experience with scalable distributed data-processing concepts.
- Created and optimized SQL queries for **data profiling, reconciliation, aggregation, backend validation, and data debugging**, identifying inconsistent records and tracing data issues across application and database layers.
- Supported implementation of logical **data models, database schemas, source-to-target mappings, and transformation rules** to ensure reliable movement of data between transactional and reporting systems.
- Implemented data validation routines for **completeness, accuracy, duplicate detection, field-level transformations, and referential integrity**, improving the quality of datasets delivered to downstream applications and reporting processes.
- Collaborated with developers, analysts, QA engineers, and business stakeholders in **Agile/Scrum teams** to understand requirements, troubleshoot distributed application issues, and deliver reliable software and data-processing functionality.
- Maintained technical documentation covering **data mappings, SQL logic, system interfaces, pipeline dependencies, debugging procedures, and operational workflows**, supporting knowledge sharing and faster investigation of production issues.

Environments: Java, Python, SQL, Apache Spark, Spark SQL, ETL, Batch Processing, Data Modeling, Oracle, MySQL, REST APIs, Distributed Systems, Linux, Jenkins, Git, Jira, Confluence, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.